REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ONE WORLD, BISHOP ALPHA AND KINDOROKO SECONDARY SCHOOLS
FIRST FORM FOUR JOINT PRE – MOCK ASSESSMENT

033/1 **BIOLOGY 1**

(For Both Private and School Candidates)

Time: 2:30 Hours **Year: 2025**

Instructions

1. This paper consists of sections A, B and C with a total of **eleven (11)** questions.
2. Answer **all** questions in sections A and B and **two (2)** questions from section C.
3. Section A carries **sixteen (16)** marks, section B **fifty four (54)** marks and section C carries **thirty (30)** marks.
4. Except for drawings which must be drawn in pencil, all writings must be in **blue** or **black** ink.
5. Cellular phones and any unauthorized materials are **not** allowed in the assessment room.
6. Write your **Name in Full** on every page of your answer booklet(s).

FIRST JOINT PRE–MOCK 2025

SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter besides the item number in the answer booklet(s) provided.

(i) Form four students at Bishop Alpha Memorial High School conducted a scientific research and wrote a very good report. What name is given to the summarized statement of their finding?

A Data interpretation
B Hypothesis
C Data collection
D Observation
E Conclusion

(ii) The alcoholic fermentation in brewery industries is not a human physiological influence because it is the result of:

A Photosynthesis process
B Anaerobic respiration by muscles
C Anaerobic respiration by fungi
D Aerobic respiration by bacteria
E Aerobic respiration by plants

(iii) The genus and species of human being can be correctly written together to bring one name which is:

A *Homo Sapiens*
B *Homo sapiens*
C *homo Sapiens*
D *homo sapiens*
E *HOMO sapiens*

(iv) What is the role of node of Ranvier in a neurone?

A To transmit the impulses away from the cell body
B To insulate the axon and speed up the transmission of impulses
C To transmit nerve impulses from one neurone to another
D To speed up the transmission of nerve impulses
E To transmit the nerve impulses towards the cell body

(v) Why is transfusion of blood from a ***group O donor*** to a ***group AB recipient*** successful?

A The recipient's blood has no antibodies
B The recipient's blood has no antigens
C The donor's blood has many antigens
D The donor's blood has many antibodies
E The recipient's blood has antigen O

(vi) Which of the following sets in the mammalian body contains smooth muscles?

A Triceps, blood vessels and urinary bladder
B Intestine, blood vessels and urinary bladder
C Intestine, triceps and blood vessels
D Intestine, triceps and urinary bladder
E Heart, triceps and urinary bladder

(vii) After drinking beer, Babymaya frequently went to the toilet to urinate. Why did this happen?

A She had more water in the body
B Alcohol relaxed her urinary bladder muscles
C Alcohol prevented re-absorption of water in her kidneys
D Urinating prevented her from getting drunk
E Alcohol allowed re-absorption of more water in her kidneys

(viii) During photosynthesis:
A Macromolecules are built from micromolecules
B Micromolecules are synthesized from macromolecules
C Heterotrophs produce their own food using sunlight
D Saprophytes use chlorophyll to manufacture their own food
E Stomata close and glucose is produced

(ix) Mr. Arip got an accident when he was slashing long grasses around his house. The accident caused bleeding through cuts. As a good neighbour, how would you help Mr. Arip before taking him to the hospital?
A Laying him on his back while the legs are raised
B Dressing his wound and bandage it firmly
C Giving him a lot of fruit juice to restore mineral salts lost
D Inducing vomiting to Mr. Arip
E Calling professional doctors to help Mr. Arip

(x) Three students from One World Secondary School conducted a project work on the characteristics of different plants. They found some seeds, each with two cotyledons and planted them. What will the students observe to the plants after such seeds germinate?
A The plants will produce cones and soft wood
B The plants will have fibrous roots and yellow flowers
C The plants will have net-like veins on their leaves
D The plants will lack vascular tissues like xylem and phloem
E The plants will have tap roots and flowers with three petals

2. Match the explanations about the theories of the origin of life in **List A** with their responses in **List B** by writing the letter of the correct response besides the item number in the answer booklet(s) provided.

LIST A	**LIST B**
(i) Life resulted from physical and chemical combination of the naturally occurring elements and molecules.	A Special creation theory
(ii) The earth has no origin; it did not have a moment of creation; hence, life has no origin.	B Cosmozoic theory
(iii) The concept that ‘life must originate from pre-existing life’.	C Naturalistic theory
(iv) The earth and all life forms on it were made in their present form by the Supreme Being (God).	D Biological theory
(v) Life arose from non-living matters on a number of distinct occasions.	E The law of biogenesis
(vi) Life on earth originated from somewhere else in the universe and brought to the earth as the readymade material.	F Abiogenesis theory
	G Steady state theory
	H Bipedalism theory

SECTION B (54 Marks)

Answer **all** questions in this section

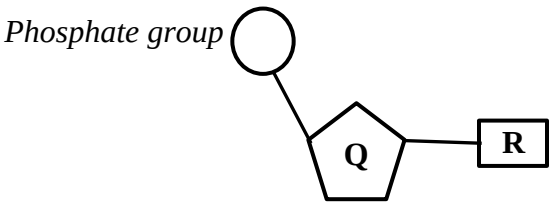
3. (a) There is big debate on why biology should not be taken as an optional subject like other optional subjects at ordinary secondary school. As an expert in biology, state five reasons to why biology should be studied by all o-level students.
   (b) Give four differences between mango cell and rod-shaped bacteria.

4. The following organisms were found in the ecosystem: Grass, Frog, Vulture, Grasshopper, Lizard and Beetle.
   (a) Construct a food web from the organisms named above.
   (b) From the food web that you have constructed in (a), which organism(s) are:
      (i) Producers
      (ii) Secondary consumers
      (iii) In the second trophic level?
   (c) (i) Which important group of organisms should be introduced in the ecosystem above to make it balanced?
      (ii) Organisms of the group named in c(i) are commonly found into two kingdoms. Mention the kingdoms.

5. (a) In order for an organism to grow well, there must be a particular kind of cell division.
      (i) Identify and give meaning of that kind of cell division.
      (ii) How is the kind of cell division in a(i) above significant? Explain four points.
   (b) Why is placenta formed soon after fertilization in humans? Provide three points.

6. (a) Briefly explain why a grasshopper would not die with its head dipped into water for some time.
   (b) With examples, distinguish between endotherms and ectotherms.
   (c) Explain four ways through which water is lost from the human body.

7. (a) Students at Kindoroko Girls' Secondary School are planning to stop cleaning the school environment for six months since they want to focus on studies only. Explain to them any four consequences of their decision.
   (b) Study the figure below and then answer the questions that follow:



      (i) What is the name of the structure illustrated above?
      (ii) State two different polymers that can be formed by joining many structures above.
      (iii) Give two differences between the polymers you have stated in (ii) above basing on the parts Q and R.

8. (a) Doctor Evance mixed a patient's sample of urine with Benedict's solution in a test tube and heated the mixture for five minutes. The colour of the mixture changed to orange.
   (i) What substance was present in the patient's urine sample?
   (ii) What would Doctor Evance conclude on the health status of the patient?
   (iii) Which organ in the patient's body might have lost its normal functioning ability?
   (b) It is common to see organisms moving from one place to another. How is this process important to the living things? Briefly explain by giving three points.

SECTION C (30 Marks)

Answer **two (2)** questions in this section

9. The brain performs a great role in controlling many activities of the human body. How is this organ adapted to perform its functions? Explain six points.

10. There is a lot of microorganisms around us which can cause harm to our health. With six points, justify how our bodies defend themselves against such microorganisms.

11. Explain three types and three significances of transpiration in plants.

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